



Class: SY

Course Name: Strength of Materials

Course Code: PCC-ME201

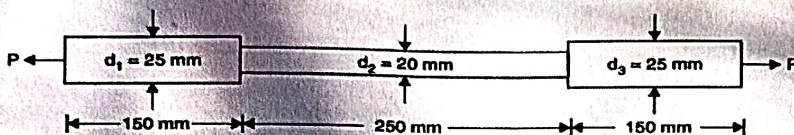
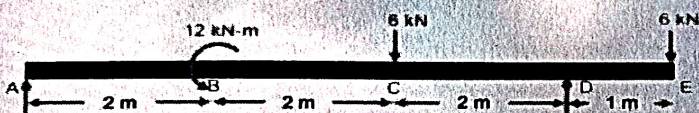
Date: 19.11.2025

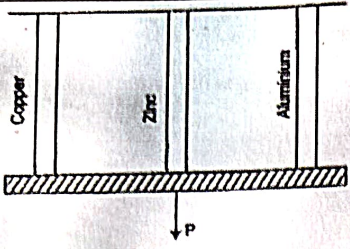
Time: 13:30 to 15:30

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
1	a) State assumption of pure bending	4	CO3	BT1
	b) Write difference in isotropic, orthotropic and anisotropic material give an example of each material	6	CO2	BT1
2.	The circular bar shown in figure is subjected to a load of 40 kN. Young's modulus of the material is 200 GPa. Total elongation of the bar is _____ mm. 	10	CO1	BT3
3 (a)	Derive Euler formula for buckling for any 2 cases	5	CO5	BT1
3 (b)	Find out the stress generated due to applied pressure "P" for hemispherical dome of radius "R" and thickness "t"	5	CO5	BT2
4	Draw shear force and bending moment diagram. 	10	CO3	BT3
5	Three bars made of copper, zinc and aluminium are of equal length and have cross-section 500, 750 and 1000 square mm respectively. They are rigidly connected at their ends. If this compound member is subjected to a longitudinal pull of 350 kN, estimate the proportional of the load carried on each rod and the induced stresses. Take the value of E for copper 1.3×10^5 N/mm², for zinc = 1.0×10^5 N/mm² and for aluminium = 0.8×10^5 N/mm².	10	CO2	BT3

				
6	A beam of length 6 m is simply supported at its ends and carries two point loads of loads of 48 kN and 40 Kn at distance of 1m and 3m respectively from the left support Find: i) Deflection under each load ii) Macium deflection and iii) The point at which maximum deflections occur Given $E=2 \times 10^5 \text{ N/mm}^2$ and $I=85 \times 10^6 \text{ mm}^4$	10	CO4	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the various concepts and principles used in mechanics of deformable materials.
CO 2	Evaluate the strains and deformation that will result due to the elastic stresses developed within the materials for simple types of loading.
CO 3	Analyze various types of stress, strains and deformations in beams, shafts, pressure vessels, etc.
CO 4	Evaluate the slope, deflections in beams and columns.
CO 5	Analyze the failure of columns under various loadings.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
End Semester Examination
Academic Year: 2025-26 (Semester I)

Class: SY	Subject: Materials Engineering	Subject Code: PCC-ME205
Date: 21.11.2024	Time: 13:30 to 15:30	Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q1	Select the material for the given component and justify your answer with a scientific reason, also explain that material in detail. a) Axe b) Engine Block	10	CO1 CO2	BT4
Q.2	Explain the Advanced material, Graphene, and Carbon Fibre material in detail with advantages, disadvantages and applications	10	CO5	BT1
Q.3	Elaborate on the iron-iron carbon diagram in detail, along with their reactions and microstructure.	10	CO2	BT1
Q.4	Write a note on the grey cast iron and white cast iron (microstructure, chemical composition and properties)	10	CO2	BT1
Q.5	Classify and explain the Bulk heat treatment process. Why the surface treatment is the first heat treatment process?	10	CO3	BT2
Q.6	Write the types of powder production processes in detail with diagrams, advantages, and limitations.	10	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Identify the various materials, viz., metals, polymers, ceramics, and composites
CO 2	Analyse the dependance of mechanical properties on the microstructure
CO 3	Select appropriate heat treatment for metals and alloys for particular application
CO 4	Acquire knowledge about powder metallurgy process and its applications.
CO 5	Acquire knowledge about advanced materials and its applications.

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded		
	Academic Year: 2025-26	Semester: I	Examination: End Term
	Class: Second Year	Branch: Mechanical Engineering	
	Date: 24/11/2025	Time: 13:30 - 15:30	
Subject: Thermodynamics		Code: PCC-ME201/ME202	Maximum Marks: 60

General Instructions:

- I. This paper contains 1 + 3 = 4 (Four) pages. Figures to right indicate full marks.
- II. Draw neat & compact sketches wherever necessary.
- III. Assume data wherever required and state them clearly.
- IV. Notations used have their usual meaning. No need to attach steam tables to the answer sheets.

Q.N.	Question	Marks	CO	BT
1.	Consider the scenario: Your mother says "The LPG gas come to end, so arrange another cylinder, else no lunch at home." As a Mechanical Engineer, do you really think the current LPG Cylinder is completely empty? Answer and explain wisely.	10	CO2	BT3
2.	Identify the relation for below conditions using =, +, -, ×, ÷, <, > . -Compared to saturated liquid, compressed liquid is characterized by: a. Pressure: $P _ P_{sat}$ at given T b. Temperature: $T _ T_{sat}$ at given P c. Specific volume: $v _ v_f$ at given P or T d. Internal Energy: $u _ u_f$ at given P or T e. Enthalpy: $h _ h_g$ at given P or T . -Explain all these parameters on the $p-v$ diagram.	5+5	CO3	BT3
3.	A football was inflated to a gauge pressure of 1 bar when the ambient temperature was 15°C. Assume that, volume of football remains constant at 2500 cm ³ . Find the amount of heat lost by the air in the football and the gauge pressure of air in the football at the stadium [Hint: Gauge pressure = Absolute pressure - Atmospheric pressure (i.e. 1.01325bar). Use absolute pressure in the all calculations.]	10	CO3	BT4
4.	Steam flows at a reate of 4 Kg/s through a steam turbine, entering the turbine at 400°C and 1000 kPa and exiting it as saturated steam at 50 kPa. Determine the power developed and isentropic efficiency of the turbine.	5	CO4	BT5
5.	An insulated box is divided in to two compartments A and B by negligibly thin partition. Compartment A initially contains 10 kg of N ₂ (MW = 25 kg/kmol, $\gamma = 1.4$) at 500 K and 500 kPa, and compartment B initially contains 2 kg of H ₂ (MW = 2 kg/kmol, $\gamma = 1.4$) at 300 K and 100 kPa. The partition is now removed and the gases are allowd to mix and reach an equilibrium state. Taking the box as the system, determine its entropy change.	10	CO4	BT4
6.	Draw the schematic of thermal power plant, explaining the its working. Also explain the cycle on which it work along with calculation of net efficiency.	5+5	CO5	BT1
7.	Diesel Engine has a compression ratio (r_k) of 14, and cut-off takes place at 6% of the stroke. Find the air standard efficiency, given as $\eta = 1 - \frac{1}{\gamma \times r_k^{\gamma-1}} \times \frac{r_c^{\gamma-1} - 1}{r_c - 1},$ where r_k and r_c is cut-off and compression ratio respectively.	5	CO5	BT5

**** ALL THE BEST ****

CO1: To impart the principles of work and energy.

CO2: To inculcate fundamentals of thermodynamics laws, concepts & principles.

CO3: To comprehend knowledge about properties of pure substance.

CO4: To comprehend knowledge about entropy, irreversibility and availability

CO5: To make familiar with various thermodynamic cycles and to apply thermodynamic concepts in various applications.

BT1: Remember, BT2: Understand, BT3: Apply, BT4: Analyze, BT5: Evaluate, BT6: Create.